Math 502
Homework #1

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Exercise 0. *In a short (less than half a page) paragraph, tell me about yourself and what you are hoping to get out of this course. For example, is there a numerical component to your thesis research? Is there a certain topic you were hoping to cover this quarter?*

Answer. My name is Gianluca Crescenzo. I'm in the mathematics masters program, and I did my undergrad at Occidental College in Los Angeles. Besides math I like rock climbing, hiking, and playing videogames. I'm taking this class because I don't have much experience or proficiency in applied math. The only applied courses I've taken during my undergrad were partial differential equations, numerical analysis, and computational mathematics.

Exercise A.1. *Let* $e = (2, -5, 3, 4)$. *Calculate the following quantities:*

$$\text{(a) } \|e\|_\infty \quad \text{(b) } \|e\|_1 \quad \text{(c) } \|e\|_2$$

Solution.

$$\begin{aligned}
\|e\|_\infty &= \max\left\{|2|, |-5|, |3|, |4|\right\} = 5. \\
\|e\|_1 &= |2| + |-5| + |3| + |4| = 14. \\
\|e\|_2 &= \sqrt{2^2 + (-5)^2 + 3^2 + 4^2} = \sqrt{4 + 25 + 9 + 16} = \sqrt{54}.
\end{aligned}$$

Exercise A.2. *Let* $A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{pmatrix}$. *Calculate the following:*

$$\text{(a) } \|A\|_\infty \quad \text{(b) } \|A\|_1 \quad \text{(c) } \|A\|_2$$

Solution. Below is the MATLAB code used to solve this exercise:

```
A = [1 2 3 4;5 6 7 8;9 10 11 12;13 14 15 16];
L_infty = max(sum(abs(A),2))
L1 = max(sum(abs(A),1))
L2 = sqrt(max(eig(A'*A)))
```

```
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L_infty =

    58

L1 =

    40

L2 =

38.6227

>>
```

Exercise A.3. *Let* $u(x) = x^2 - x$ *be a function defined on* $0 \leqslant x \leqslant 1$*. Calculate the following quantities:*

$$(a)\ \|u\|_\infty \quad (b)\ \|u\|_1 \quad (c)\ \|u\|_2$$

Solution. The infinity norm of $u(x) = x^2 - x$ is $\|u\|_\infty = \sup\{|u(x)| \mid x \in [0,1]\} = 1/2$. The one norm and two norm can be seen from the following MATLAB code:

```
syms x
u = x^2 - x;
L1_exact = int(abs(u), x, 0, 1)
L2_exact = sqrt(int(abs(u)^2, x, 0, 1))
```

```
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L1_exact =

1/6


L2_exact =

30^(1/2)/30

>>
```

Exercise A.4. *Let U be the grid function defined by $U_i = x_i^2 - x_i$, where $x_1 = 0.2$, $x_2 = 0.4$, $x_3 = 0.6$, $x_4 = 0.8$, $x_5 = 1.0$. Calculate the following quantities:*

$$(a)\ \|U\|_\infty \quad (b)\ \|U\|_1 \quad (c)\ \|U\|_2$$

Solution. Below is the MATLAB code used to solve this exercise:

```
syms x
u = x^2 - x;
h = [0.2 0.4 0.6 0.8 1.0];
U = double(subs(u, x, h)) ;
Uinf = max(abs(U))
U1 = (0.2) * sum(abs(U))
U2 = sqrt((0.2) * sum(abs(U).^2))
```

```
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Uinf =

    0.2400


U1 =

    0.1600


U2 =

    0.1824

>>
```

Exercise 1.2.

(1) *Use the method of undetermined coefficients to set up the 5×5 Vandermonde system that would determine a fourth-order accurate finite difference approximation to $u''(x)$ based on 5 equally spaced points:*

$$u''(x) = c_{-2}u(x-2h) + c_{-1}u(x-h) + c_0u(x) + c_1u(x+h)c_2u(x+2h)$$

(2) *Compute the coefficients using the MATLAB code `fdstencil.m`, and check that they satisfy the system you determined in part (1).*

(3) *Test this finite difference approximate* $u''(1)$ *for* $u(x) = \sin(2x)$ *with values of* h *from the array* `hvals = logspace(-1,-4,13)`. *Make a table of the error vs* h *for several values of* h *and compare against the predicted error from the leading term of the expression printed by* `fdstencil.m`. *Also produce a log-log plot of the absolute value of the error vs* h.

Solution.

(1) Taylor expanding $u(x-2h)$, $u(x-h)$, $u(x+h)$, and $u(x+2h)$ gives:

$$\begin{aligned}
u''(x) = {} & c_{-2}\left(u(x) + (-2h)u'(x) + \frac{(-2h)^2}{2!}u''(x) + \frac{(-2h)^3}{3!}u'''(x) + \frac{(-2h)^4}{4!}u''''(x) + O(h^5)\right) \\
& + c_{-1}\left(u(x) + (-h)u'(x) + \frac{(-h)^2}{2!}u''(x) + \frac{(-h)^3}{3!}u'''(x) + \frac{(-2h)^4}{4!}u''''(x) + O(h^5)\right) \\
& + c_0 u(x) \\
& + c_1\left(u(x) + (h)u'(x) + \frac{(h)^2}{2!}u''(x) + \frac{(h)^3}{3!}u'''(x) + \frac{(-2h)^4}{4!}u''''(x) + O(h^5)\right) \\
& + c_2\left(u(x) + (2h)u'(x) + \frac{(2h)^2}{2!}u''(x) + \frac{(2h)^3}{3!}u'''(x) + \frac{(-2h)^4}{4!}u''''(x) + O(h^5)\right).
\end{aligned}$$

Expanding each of the constants and factoring out the respective derivatives of $u(x)$ gives:

$$\begin{aligned}
u''(x) = {} & (c_{-2} + c_{-1} + c_0 + c_1 + c_2)u(x) \\
& + \big(c_{-2}(-2h) + c_{-1}(-h) + c_1(h) + c_2(2h)\big)u'(x) \\
& + \left(c_{-2}\frac{(-2h)^2}{2!} + c_{-1}\frac{(-h)^2}{2!} + c_1\frac{(h)^2}{2!} + c_2\frac{(2h)^2}{2!}\right)u''(x) \\
& + \left(c_{-2}\frac{(-2h)^3}{3!} + c_{-1}\frac{(-h)^3}{3!} + c_1\frac{(h)^3}{3!} + c_2\frac{(2h)^3}{3!}\right)u'''(x) \\
& + \left(c_{-2}\frac{(-2h)^4}{4!} + c_{-1}\frac{(-h)^4}{4!} + c_1\frac{(h)^4}{4!} + c_2\frac{(2h)^4}{4!}\right)u'''(x) \\
& + O(h^5).
\end{aligned}$$

We can rewrite this as the following system of equations:

$$\begin{aligned}
c_{-2} + c_{-1} + c_0 + c_1 + c_2 &= 0 \\
c_{-2}(-2h) + c_{-1}(-h) + c_1(h) + c_2(2h) &= 0 \\
c_{-2}\frac{(-2h)^2}{2!} + c_{-1}\frac{(-h)^2}{2!} + c_1\frac{(h)^2}{2!} + c_2\frac{(2h)^2}{2!} &= 1 \\
c_{-2}\frac{(-2h)^3}{3!} + c_{-1}\frac{(-h)^3}{3!} + c_1\frac{(h)^3}{3!} + c_2\frac{(2h)^3}{3!} &= 0 \\
c_{-2}\frac{(-2h)^4}{4!} + c_{-1}\frac{(-h)^4}{4!} + c_1\frac{(h)^4}{4!} + c_2\frac{(2h)^4}{4!} &= 0.
\end{aligned}$$

Hence we've obtained the following matrix equation:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ -2h & -h & 0 & h & 2h \\ \frac{(-2h)^2}{2!} & \frac{(-h)^2}{2!} & 0 & \frac{h^2}{2!} & \frac{(2h)^2}{2!} \\ \frac{(-2h)^3}{3!} & \frac{(-h)^3}{3!} & 0 & \frac{h^3}{3!} & \frac{(2h)^3}{3!} \\ \frac{(-2h)^4}{4!} & \frac{(-h)^4}{4!} & 0 & \frac{h^4}{4!} & \frac{(2h)^4}{4!} \end{pmatrix} \begin{pmatrix} c_{-2} \\ c_{-1} \\ c_0 \\ c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

Factoring an h^2 out of the third row and simplifying yields:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ -2h & -h & 0 & h & 2h \\ 2 & \frac{1}{2} & 0 & \frac{1}{2} & 2 \\ -\frac{4h^3}{3} & -\frac{h^3}{6} & 0 & \frac{h^3}{6} & \frac{4h^3}{3} \\ \frac{2h^4}{3} & \frac{h^4}{24} & 0 & \frac{h^4}{24} & \frac{2h^4}{3} \end{pmatrix} \begin{pmatrix} c_{-2} \\ c_{-1} \\ c_0 \\ c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \frac{1}{h^2} \\ 0 \\ 0 \end{pmatrix}.$$

(2) The following code row reduces the the augmented matrix obtained from the above matrix equation, as well as prints the output of `fdstencil(2,-2:2)`.

```
syms h

A = [       1,          1, 1,          1,          1,      0;
         -2*h,         -h, 0,          h,        2*h,      0;
            2,        1/2, 0,        1/2,          2, 1/h^2;
   -(4*h^3)/3, -(h^3)/6, 0,   (h^3)/6, (4*h^3)/3,      0;
    (2*h^4)/3, (h^4)/24, 0, (h^4)/24, (2*h^4)/3,      0];

disp(rref(A))
fdstencil(2,-2:2)
```

```
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[1, 0, 0, 0, 0, -1/(12*h^2)]
[0, 1, 0, 0, 0,   4/(3*h^2)]
[0, 0, 1, 0, 0,  -5/(2*h^2)]
[0, 0, 0, 1, 0,   4/(3*h^2)]
[0, 0, 0, 0, 1, -1/(12*h^2)]


The derivative u^(2) of u at x0 is approximated by

 1/h^2 * [
 -8.333333333333333e-02 * u(x0-2*h) +
  1.333333333333333e+00 * u(x0-1*h) +
 -2.500000000000000e+00 * u(x0) +
  1.333333333333333e+00 * u(x0+1*h) +
 -8.333333333333333e-02 * u(x0+2*h) ]

For smooth u,
 Error = 0 * h^3*u^(5) + -0.0111111 * h^4*u^(6) + ...

>>
```

Hence the coefficients obtained by `fdstencil.m` satisfy the system we determined in part (1).

(3) The following MATLAB code and output addresses the question:

```
% u(x) = sin(2x), approximate u''(1):
uptrue = -4*sin(2);
%hvals = [1e-1 5e-2 1e-2 5e-3 1e-3];
hvals = logspace(-1, -4, 13);
E5u = [];
% table headings:
disp(' ')
disp(' h D3u E3u')
for i=1:length(hvals)
h = hvals(i);
% approximation to u''(1) using fdcoeffF:
xpts = 1 + h*(-2:2)';
D5u = fdcoeffF(2,1,xpts) * sin(2*xpts) ;
% error:
E5u(i) = D5u - uptrue;
% print line of table:
```

```
disp(sprintf('%13.4e %13.4e %13.4e',...
h,D5u,E5u(i)))
end
% plot errors:
clf
loglog(hvals,abs(E5u),'o-')
axis([5e-7 .2 1e-12 1])
```

```
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   h              D3u            E3u
   1.0000e-01    -3.6371e+00     6.4431e-05
   5.6234e-02    -3.6372e+00     6.4588e-06
   3.1623e-02    -3.6372e+00     6.4638e-07
   1.7783e-02    -3.6372e+00     6.4655e-08
   1.0000e-02    -3.6372e+00     6.4670e-09
   5.6234e-03    -3.6372e+00     6.4623e-10
   3.1623e-03    -3.6372e+00     6.4150e-11
   1.7783e-03    -3.6372e+00    -6.6817e-11
   1.0000e-03    -3.6372e+00    -1.8323e-10
   5.6234e-04    -3.6372e+00    -1.8130e-09
   3.1623e-04    -3.6372e+00     4.2406e-09
   1.7783e-04    -3.6372e+00     6.1032e-09
   1.0000e-04    -3.6372e+00    -3.8600e-08
>>
```

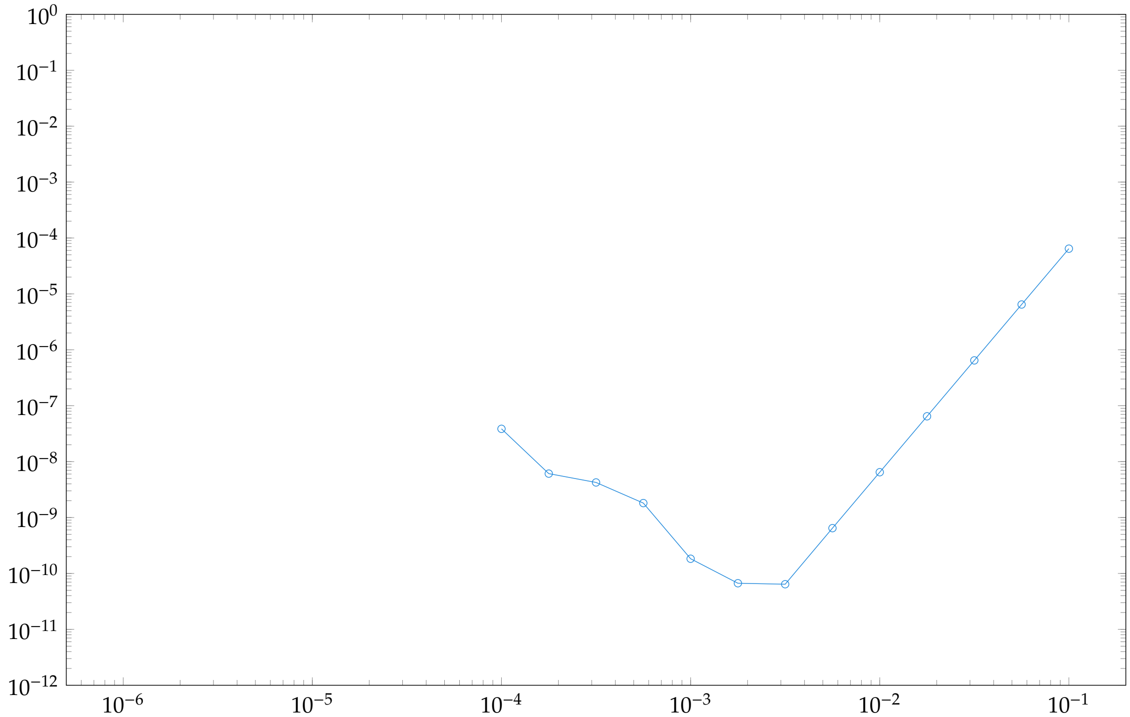

10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
10^{-10}
10^{-11}
10^{-12}
10^{-6}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}